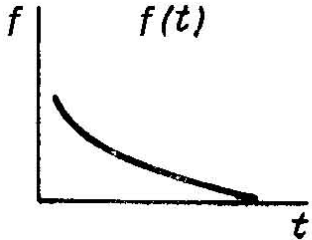
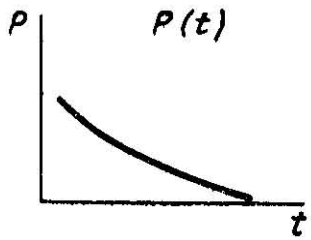
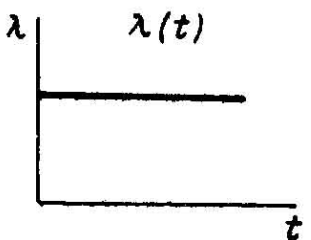
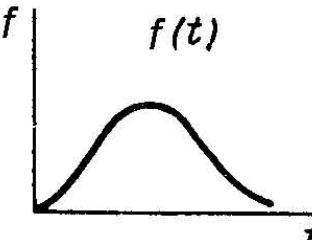
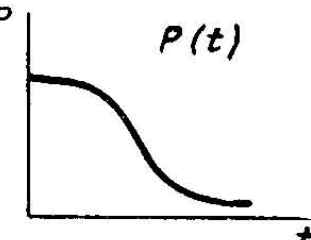
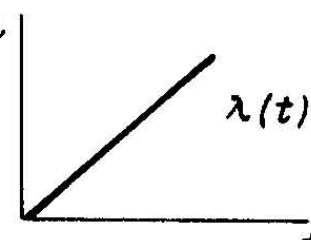
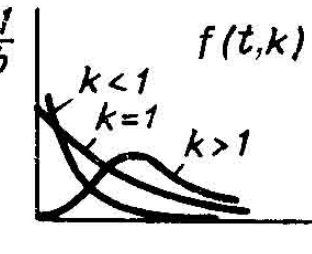
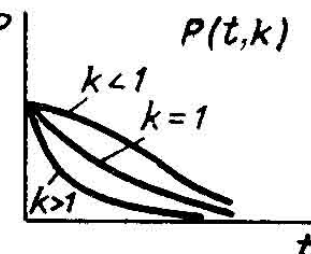
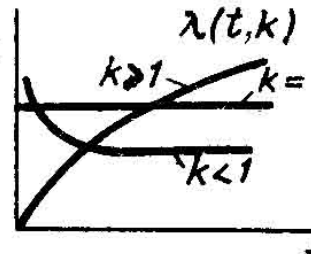
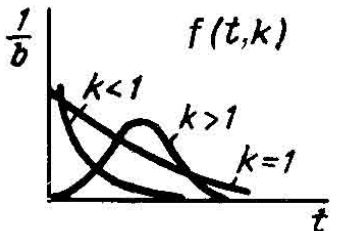
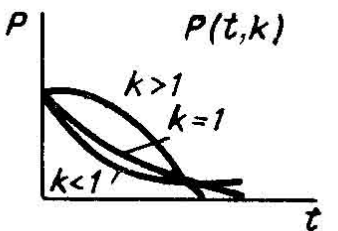
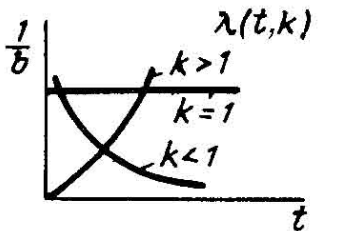
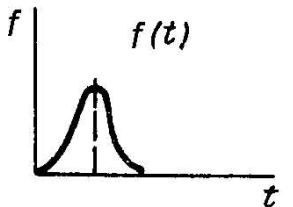
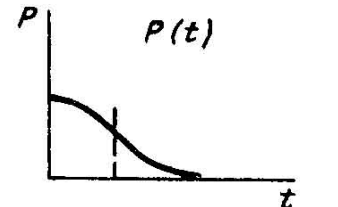
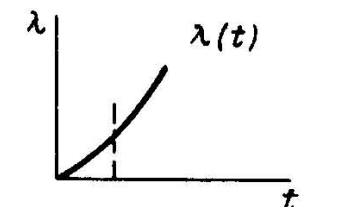
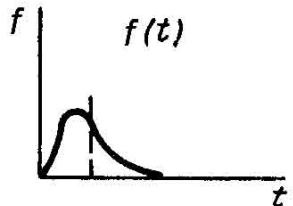
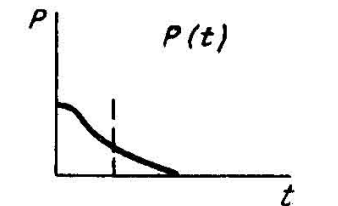
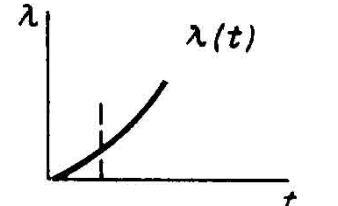


Закон распределения времени до отказа		Основные соотношения для количественных характеристик надежности		
Наименование	Плотность распределения	Вероятность безотказной работы	Интенсивность отказов	Средняя наработка до первого отказа
Экспоненциальный	$f(t) = \lambda e^{-\lambda t}$ 	$P(t) = e^{-\lambda t}$ 	$\lambda(t) = \text{const}$ 	$T_1 = \frac{1}{\lambda}$
Релея	$f(t) = \frac{t}{\sigma^2} e^{-\frac{t^2}{2\sigma^2}}$ 	$P(t) = e^{-\frac{t^2}{2\sigma^2}}$ 	$\lambda(t) = \frac{t}{\sigma^2}$ 	$T_1 = \sqrt{\frac{\pi}{2}} \sigma$
Гамма (при k целом)	$f(t) = \lambda_0 \frac{(\lambda_0 t)^{k-1}}{(k-1)!} e^{-\lambda_0 t}$ 	$P(t) = e^{-\lambda_0 t} \sum_{i=0}^{k-1} \frac{(\lambda_0 t)^i}{i!}$ 	$\lambda(t) = \frac{\lambda_0 (\lambda_0 t)^{k-1}}{(k-1)! \sum_{i=0}^{k-1} \frac{(\lambda_0 t)^i}{i!}}$ 	$T_1 = \frac{k}{\lambda_0}$

Вейбулла	$f(t) = \lambda_0 k t^{k-1} e^{-\lambda_0 t^k}$ 	$P(t) = e^{-\lambda_0 t^k}$ 	$\lambda(t) = \lambda_0 k t^{k-1}$ 	$T_1 = \frac{\Gamma\left(\frac{1}{k} + 1\right)}{\lambda_0^{\frac{1}{k}}}$
Усеченный нормальный	$f(t) = \frac{1}{F\left(\frac{T_0}{\sigma}\right) \sigma \sqrt{2\pi}} e^{-\frac{(t-T_0)^2}{2\sigma^2}}$ 	$P(t) = \frac{F\left(\frac{T_0 - t}{\sigma}\right)}{F\left(\frac{T_0}{\sigma}\right)}$ 	$\lambda(t) = \frac{e^{-\frac{(t-T_0)^2}{2\sigma^2}}}{\sqrt{2\pi}\sigma F\left(\frac{T_0 - t}{\sigma}\right)}$ 	$T_1 = T_0 + \frac{\sigma}{\sqrt{2\pi} F\left(\frac{T_0}{\sigma}\right)} e^{-\frac{T_0^2}{2\sigma^2}}$
Логарифмически-нормальный	$f(t) = \frac{1}{\sigma t \sqrt{2\pi}} e^{-\frac{1}{2} \left(\frac{\ln t - \mu}{\sigma}\right)^2}$ 	$P(t) = \frac{1}{2} + \Phi\left(\frac{\mu - \ln t}{\sigma}\right)$ 	$\lambda(t) = \frac{1}{\sqrt{2\pi}\sigma t} \frac{e^{-\frac{1}{2} \left(\frac{\ln t - \mu}{\sigma}\right)^2}}{0.5 + \Phi\left(\frac{\mu - \ln t}{\sigma}\right)}$ 	$T_1 = \frac{1}{\sigma \sqrt{2\pi}} \int_0^\infty e^{-\frac{(\ln t - \mu)^2}{2\sigma^2}} dt$